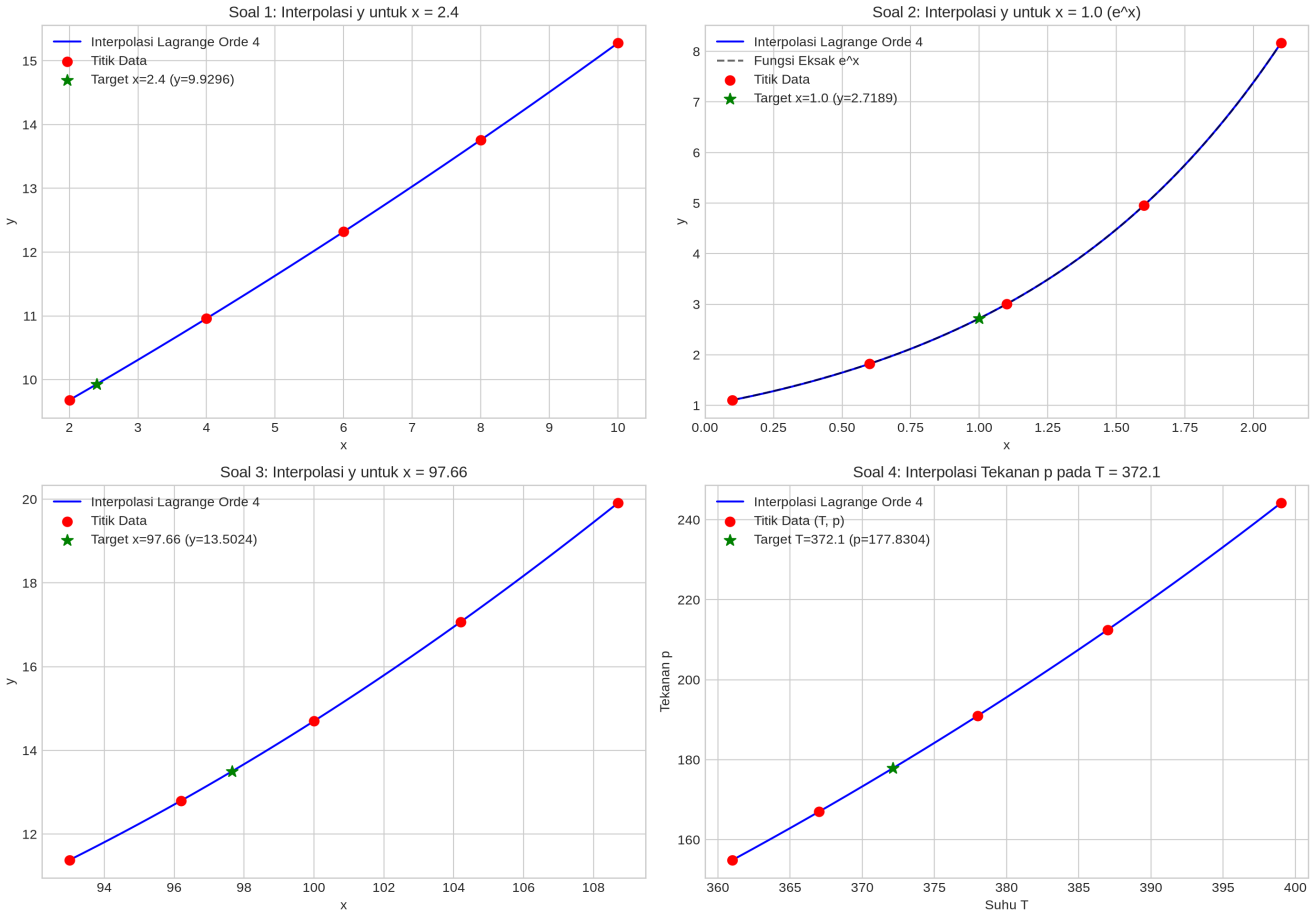
Laporan Tugas Komputasi Numerik: Pemecahan Masalah Metode Interpolasi

Mata Kuliah: Komputasi Numerik / Kelas A
Nama Mahasiswa: Nabil Aulia Dika (Dezz)
NPM: 250810701100090
Topik: Bab VI Interpolasi (Latihan 6.6 Soal 1 s.d. 4)

1. Ringkasan Hasil Perhitungan

No	Soal Target	Linier (Orde 1)	Kuadratik (Orde 2)	Lagrange (Orde 4)	Analisis / Keterangan
1	x = 2.4	9.9360	9.9296	9.9296	Polinomial orde 2 murni
2	x = 1.0 (e^x)	2.7678 (galat 1.82%)	2.7064 (galat 0.43%)	2.7189 (galat 0.02%)	Eksak e^1 = 2.718282
3	x = 97.66	13.5300	13.5025	13.5024	Kuadratik mengapit akurat
4	T = 372.1	178.1273	177.8157	177.8304	Data tekanan non-linier

2. Grafik Hasil Interpolasi Seluruh Soal



3. Ringkasan Perhitungan Matematis Tiap Soal

```
SOAL 1: x = 2.4 (Data x=[2, 4, 6, 8, 10], y=[9.68, 10.96, 12.32, 13.76, 15.28])
- Linier (x=2, 4): f1(2.4) = 9.68 + (1.28 / 2) * (0.4) = 9.9360
```

- Kuadratik: b0=9.68, b1=0.64, b2=0.01 -> f2(2.4) = 9.68 + 0.256 - 0.0064 = 9.9296

SOAL 2: x = 1.0 (Data x=[0.1, 0.6, 1.1, 1.6, 2.1], y=[1.1052, 1.8221, 3.0042, 4.9530, 8.1662])
- Eksak: e^1 = 2.718282
- Linier (x=0.6, 1.1): f1(1.0) = 1.8221 + (1.1821 / 0.5) * 0.4 = 2.7678 (Galat: 1.82%)
- Kuadratik: b0=1.8221, b1=2.3642, b2=1.5334 -> f2(1.0) = 2.7064 (Galat: 0.43%)
- Lagrange Orde 4: f4(1.0) = 2.718917 (Galat: 0.023%)

SOAL 3: x = 97.66 (Data x=[93, 96.2, 100, 104.2, 108.7], y=[11.38, 12.80, 14.70, 17.07, 19.91])
- Linier (x=96.2, 100): f1(97.66) = 12.80 + (1.90 / 3.8) * 1.46 = 13.5300
- Kuadratik: b0=12.80, b1=0.50, b2=0.008036 -> f2(97.66) = 13.5025

SOAL 4: T = 372.1 (Data T=[361, 367, 378, 387, 399], p=[154.9, 167.0, 191.0, 212.5, 244.2])
- Linier (T=367, 378): f1(372.1) = 167.0 + (24.0 / 11) * 5.1 = 178.1273
- Kuadratik: b0=167.0, b1=2.1818, b2=0.01035 -> f2(372.1) = 177.8157